

Route decision: Port of Shanghai → Port of Rotterdam

PORT-SHANGHAI → PORT-ROTTERDAM · mode sea (STRICT) · cargo general · priority normal · scenarios: RED_SEA_CONFLICT · planned 2026-09-29 13:47 UTC

Recommendation: BALANCED, SEA via Malacca, Capegood. Expected arrival in **742 h** (30.9 days); plan for 754 h (p85), worst case 771 h (p95). Cost: freight CN¥25,875. Peak leg risk 8%.

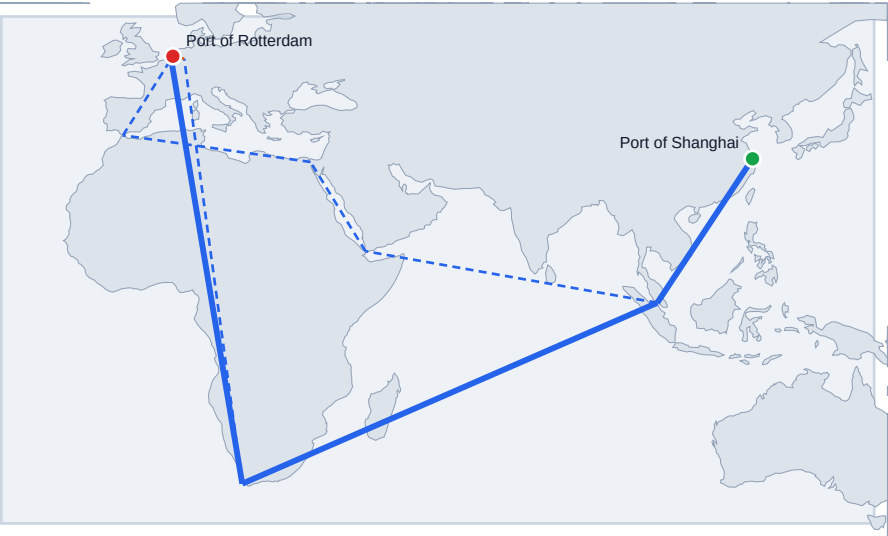
Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Risk
FASTEST	SEA	Malacca, Babel, Suez, Gibral	592	729	834	886	CN¥19,004	85%
SAFEST	SEA	Malacca, Capegood	677	742	753	770	CN¥26,876	8%
BALANCED	SEA	Malacca, Capegood	678	742	754	771	CN¥25,875	8%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: SEA via Malacca, Babel, Suez, Gibral

SEA via Malacca, Babel, Suez, Gibral; expected 728.9 h (p85 833.5 h, p95 885.8 h), CN¥19,004 freight, 0 handoffs. 12.8 h faster than the next option. CN¥6,869 cheaper (freight) than the lowest-cost alternative. Avoids Capegood. Includes 72.0 h announced scenario delay.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Strait of Malacca	3,810	108.9	+26.6	CN¥3,842	0%	
2	SEA	Bab el-Mandeb Strait	6,585	260.1	+144.2	CN¥6,639	85%	SCENARIO: Regional conflict escalation. Vessels rerouting via Cape of Good Hope for risk mitigation.
3	SEA	Suez Canal	2,302	65.8	+28.5	CN¥2,321	0%	
4	SEA	Strait of Gibraltar	3,548	101.4	+20.6	CN¥3,577	0%	
5	SEA	Port of Rotterdam	1,945	55.6	+55.3	CN¥1,961	18%	LIVE_NEWS: Temporary transformer in Europoort relieves grid congestion - Port of Rotterdam

Transit 519.7 h + transfers 0.0 h + announced scenario delay 72.0 h = plan 591.7 h. Model delay buffer p50/p85/p95: +137.2 / +275.2 / +364.2 h. Route band p50 728.9 h, p85 833.5 h, p95 885.8 h (independent legs); 955.9 h if all legs are late together. Costs: linehaul CN¥18,340, transfers CN¥0.00, disruption surcharge CN¥663.89.

Why the slowest leg (CHOKE-MALACCA → CHOKE-BABEL, p85 216.2 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: SEA via Malacca, Capegood

SEA via Malacca, Capegood; expected 741.7 h (p85 753.3 h, p95 770.0 h), CN¥26,876 freight, 2 handoffs. 12.8 h slower than FASTEST. CN¥7,871 more (freight) than FASTEST. Avoids Babel, Gibral, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Strait of Malacca	3,810	108.9	+26.6	CN¥3,842	0%	
2	SEA	Cape of Good Hope	9,573	273.5	+27.5	CN¥9,652	0%	
3	SEA	Duisburg Logport	9,608	274.5	+27.5	CN¥9,687	0%	
4	handoff	Duisburg Logport		4.0	+0.0	CN¥672.16	3%	
5	ROAD	Port of Rotterdam	166	2.4	+3.6	CN¥1,343	0%	
6	handoff	Port of Rotterdam		14.0	+0.0	CN¥1,680	8%	

Transit 659.3 h + transfers 18.0 h + announced scenario delay 0.0 h = plan 677.3 h. Model delay buffer p50/p85/p95: +64.4 / +85.3 / +115.3 h. Route band p50 741.7 h, p85 753.3 h, p95 770.0 h (independent legs); 792.6 h if all legs are late together. Costs: linehaul CN¥24,524, transfers CN¥2,353, disruption surcharge CN¥0.00.

Why the slowest leg (CHOKE-MALACCA → CHOKE-CAPEGOOD, p85 27.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: SEA via Malacca, Capegood

SEA via Malacca, Capegood; expected 742.4 h (p85 753.9 h, p95 770.6 h), CN¥25,875 freight, 2 handoffs. 13.5 h slower than FASTEST. CN¥6,869 more (freight) than FASTEST. Avoids Babel, Gibral, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Strait of Malacca	3,810	108.9	+26.6	CN¥3,842	0%	
2	SEA	Cape of Good Hope	9,573	273.5	+27.5	CN¥9,652	0%	
3	SEA	Rotterdam Distripark	9,700	277.1	+27.5	CN¥9,779	0%	
4	handoff	Rotterdam Distripark		4.0	+0.0	CN¥672.16	3%	
5	ROAD	Port of Rotterdam	31	0.5	+3.6	CN¥249.64	0%	
6	handoff	Port of Rotterdam		14.0	+0.0	CN¥1,680	8%	

Transit 660.0 h + transfers 18.0 h + announced scenario delay 0.0 h = plan 678.0 h. Model delay buffer p50/p85/p95: +64.4 / +85.3 / +115.3 h. Route band p50 742.4 h, p85 753.9 h, p95 770.6 h (independent legs); 793.3 h if all legs are late together. Costs: linehaul CN¥23,523, transfers CN¥2,353, disruption surcharge CN¥0.00.

Why the slowest leg (CHOKE-MALACCA → CHOK-CAPEGOOD, p85 27.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.